LINUX ASSIGNMENT

Name : Prathamesh Shingate

---

1. ls :

Lists files and directories in the current directory.

```
prathamesh@Prathamesh-Shingate:~$ ls
'Account Settings _ Spreezy Slack.pdf'  Desktop  Documents  Downloads  Music  Pictures  Public  snap  Templates  Videos
prathamesh@Prathamesh-Shingate:~$
```

2. ls -al :

Lists all files and directories, including hidden files, in a detailed format.
It shows permissions, owner, group, file size, and last modified time.

Explanation of Options:

- `-a` → Shows **all files**, including hidden files
- `-l` → Shows files in **long listing format** with details

Why it is used:
This command is used when a user needs **complete visibility** of directory contents, including hidden configuration files and permission details.

```
prathamesh@Prathamesh-Shingate:~$ ls -al
total 196
drwxr-x--- 23 prathamesh prathamesh  4096 Jan 11 19:28  .
drwxr-xr-x  3 root       root        4096 Jan  2 14:37  ..
-rw-rw-r--  1 prathamesh prathamesh 72983 Jan  5 21:50 'Account Settings _ Spreezy Slack.pdf'
-rw-------  1 prathamesh prathamesh  7606 Jan 13 10:58  .bash_history
-rw-r--r--  1 prathamesh prathamesh   220 Mar 31  2024  .bash_logout
-rw-r--r--  1 prathamesh prathamesh  3771 Mar 31  2024  .bashrc
drwx------ 21 prathamesh prathamesh  4096 Jan 13 19:51  .cache
drwxr-xr-x  3 prathamesh prathamesh  4096 Jan 10 23:08  .cert
drwx------ 22 prathamesh prathamesh  4096 Jan 13 19:51  .config
drwxr-xr-x  3 prathamesh prathamesh  4096 Jan 10 20:49  Desktop
drwxr-xr-x  2 prathamesh prathamesh  4096 Jan 11 21:10  Documents
drwxrwxr-x  3 prathamesh prathamesh  4096 Jan  7 22:18  .dotnet
drwxr-xr-x  3 prathamesh prathamesh  4096 Jan 13 19:38  Downloads
-rw-rw-r--  1 prathamesh prathamesh   192 Jan  2 22:19  .gitconfig
drwx------  2 prathamesh prathamesh  4096 Jan 12 23:56  .gnupg
drwxrwxr-x  4 prathamesh prathamesh  4096 Jan  9 13:58  .java
drwxrwxr-x  3 prathamesh prathamesh  4096 Jan  9 14:21  .junie
-rw-------  1 prathamesh prathamesh    20 Jan 11 19:28  .lesshst
drwx------  4 prathamesh prathamesh  4096 Jan  2 14:37  .local
drwxr-xr-x  2 prathamesh prathamesh  4096 Jan  2 14:37  Music
drwxr-xr-x  3 prathamesh prathamesh  4096 Jan  2 22:38  Pictures
drwx------  3 prathamesh docker      4096 Jan  2 22:05  .pki
-rw-r--r--  1 prathamesh prathamesh   807 Mar 31  2024  .profile
drwxr-xr-x  2 prathamesh prathamesh  4096 Jan  2 14:37  Public
drwxrwxr-x  2 prathamesh prathamesh  4096 Jan 10 00:25  .redhat
drwx------ 11 prathamesh prathamesh  4096 Jan  9 13:58  snap
drwx------  2 prathamesh prathamesh  4096 Jan  8 01:16  .ssh
-rw-r--r--  1 prathamesh prathamesh     0 Jan  2 14:40  .sudo_as_admin_successful
drwxr-xr-x  2 prathamesh prathamesh  4096 Jan  2 14:37  Templates
drwxr-xr-x  2 prathamesh prathamesh  4096 Jan  2 14:37  Videos
drwxrwxr-x  4 prathamesh docker      4096 Jan  2 22:05  .vscode
-rw-rw-r--  1 prathamesh docker       165 Jan  2 22:15  .wget-hsts
```

3. mkdir :

Creates a new directory for organizing files.

Command : **mkdir <directory-name>**

```
prathamesh@Prathamesh-Shingate:~$ mkdir linux
prathamesh@Prathamesh-Shingate:~$ ls
'Account Settings _ Spreezy Slack.pdf'  Desktop  Documents  Downloads  linux  Music  Pictures  Public  snap  Templates  Videos
prathamesh@Prathamesh-Shingate:~$
```

4. cd :

Changes the current directory to the specified path.

Command : **cd <path>**

```
prathamesh@Prathamesh-Shingate:~$ cd linux
prathamesh@Prathamesh-Shingate:~/linux$
```

5. touch :

Creates an **empty file** . Commonly used to quickly create files for practice.

Command : **touch <file-name>**

```
prathamesh@Prathamesh-Shingate:~/linux$ touch file1.txt
prathamesh@Prathamesh-Shingate:~/linux$ ls
file1.txt
prathamesh@Prathamesh-Shingate:~/linux$
```

6. cat :

Displays the entire content of a file in the terminal. Best suited for viewing small text files.

Command : **cat <file-name>**

```
prathamesh@Prathamesh-Shingate:~/linux$ cat file1.txt
This is silva from dubai
Welcome to dubai !

prathamesh@Prathamesh-Shingate:~/linux$
```

7. pwd(Present working directory) :

Displays the full path of the current working directory.
Helps users confirm their exact location in the filesystem.

Command : **pwd**

```
prathamesh@Prathamesh-Shingate:~/linux$ pwd
/home/prathamesh/linux
prathamesh@Prathamesh-Shingate:~/linux$
```

8. cp :

Copies files or directories from one location to another.

Command : **cp <source> <destination>**

```
prathamesh@Prathamesh-Shingate:~/linux$ ls
backup  file1.txt
prathamesh@Prathamesh-Shingate:~/linux$ cp file1.txt backup
prathamesh@Prathamesh-Shingate:~/linux$ cd  backup
prathamesh@Prathamesh-Shingate:~/linux/backup$ ls
file1.txt
prathamesh@Prathamesh-Shingate:~/linux/backup$
```

9. mv :

The mv command is used to move files or directories from one location to another.
It is also used to **rename files or directories** within the filesystem**.**

Command **: mv <source> <destination>**

rename :

```
prathamesh@Prathamesh-Shingate:~/linux$ touch new-1.txt
prathamesh@Prathamesh-Shingate:~/linux$ ls
backup  file1.txt  new-1.txt
prathamesh@Prathamesh-Shingate:~/linux$ mv new-1.txt new1.txt
prathamesh@Prathamesh-Shingate:~/linux$ ls
backup  file1.txt  new1.txt
prathamesh@Prathamesh-Shingate:~/linux$
```

move :

```
prathamesh@Prathamesh-Shingate:~/linux$ mv new1.txt backup
prathamesh@Prathamesh-Shingate:~/linux$ cd backup
prathamesh@Prathamesh-Shingate:~/linux/backup$ ls
new1.txt
prathamesh@Prathamesh-Shingate:~/linux/backup$
```

10. rm -r :

Deletes files or directories permanently.
The -r option allows recursive deletion of folders.

Command : **rm -r <file/folder>**

```
prathamesh@Prathamesh-Shingate:~/linux/backup$ touch new1.txt
prathamesh@Prathamesh-Shingate:~/linux/backup$ ls
new1.txt
prathamesh@Prathamesh-Shingate:~/linux/backup$ cd ..
prathamesh@Prathamesh-Shingate:~/linux$ rm -r backup
prathamesh@Prathamesh-Shingate:~/linux$
```

11. cd . .

The **cd ..** command changes the current directory to its parent directory. It is commonly used to move one level up in the directory structure while navigating the filesystem.

Command : **cd . .**

```
prathamesh@Prathamesh-Shingate:~/linux$ cd Backup
prathamesh@Prathamesh-Shingate:~/linux/Backup$ cd ..
prathamesh@Prathamesh-Shingate:~/linux$
```

12. less :

The less command is used to view the contents of a file in an interactive and scrollable format.

Command : **less <file-name>**

eg → less file2.txt

```
prathamesh@Prathamesh-Shingate:~/linux$ vi file2.txt
prathamesh@Prathamesh-Shingate:~/linux$ less file2.txt
```

```
prathamesh@Prathamesh-Shingate: ~/linux
A 50-line paragraph is
extremely long, essentially an essay or short story, requiring a central theme like life's journey to sustain it; it would start with an introduction, develop ideas with multiple supp
orting sentences (like describing bustling city mornings or simple acts of kindness), and conclude with a summary, all flowing logically to meet the line count, though typical paragra
phs are much shorter, focusing on a single idea.
Here's a demonstration of a long paragraph that could reach 50 lines (though formatting varies):
The world, a tapestry woven from threads of diverse experiences, unfolds each day with a silent promise of novelty, a whisper of the extraordinary hidden within the mundane. From the
first blush of dawn painting the eastern sky to the final twinkling of stars, life moves in rhythms both grand and subtle, a symphony of human endeavor and natural wonder. Consider th
e sprawling metropolis, a hive of ceaseless activity where early risers become a human river flowing to work, a vibrant mosaic of cultures, ambitions, and dreams. The air buzzes with
a thousand conversations, the aroma of street food, and the distant hum of traffic, a concrete jungle alive with interconnected stories. Yet, amidst this chaos, pockets of serenity ex
ist, a quiet park where an elderly couple shares a bench, or a lone artist captures the fleeting beauty of a sunset on canvas.
Then there are the profound moments of human connection, those small, unassuming acts of kindness that ripple outward, transforming a stranger's difficult day into one of hope. A shar
ed smile, a held door, a patient listening ear—these gestures, seemingly insignificant, form the bedrock of our shared humanity, reminding us we are not islands. Think of the child's
wonder discovering a ladybug on a leaf, or the quiet satisfaction of a gardener tending to fragile blooms, each a testament to life's enduring magic. Our journey is also marked by anc
ient whispers from civilizations past, echoed in monumental architecture, stories etched in stone that connect us to millennia of human struggle and triumph, like the enduring mystiqu
e of the Himalayas or the vibrant festivals that celebrate shared heritage. Even in quiet moments of solitude, looking up at the vast, star-dusted expanse, one feels a deep connection
 to something immense, a universal narrative that binds us all. This vast, intricate narrative, encompassing bustling cities, quiet acts of love, and the enduring legacy of history, i
s the essence of our shared existence, a continuous story waiting for each of us to write our own verse. It's a story of resilience, of fleeting beauty, and of the profound, often und
erstated, connections that make life truly meaningful, a ceaseless cycle of beginnings and endings, of joy and sorrow, all woven into the grand, ongoing story of the world.

~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
```

13. head :

The head command displays the first few lines of a file, with a default of 10 lines. It is commonly used to quickly check the beginning or header content of a file.

We can specify the number as well like for eg →

head -5 file2.txt Displays the first 5 lines of the text file.

Command : **head <file-name>**

```
prathamesh@Prathamesh-Shingate:~/linux$ head file2.txt
A 50-line paragraph is
extremely long, essentially an essay or short story, requiring a central theme like life's journey to sustain it; it would start with an introduction, develop ideas with multiple supp
orting sentences (like describing bustling city mornings or simple acts of kindness), and conclude with a summary, all flowing logically to meet the line count, though typical paragra
phs are much shorter, focusing on a single idea.
Here's a demonstration of a long paragraph that could reach 50 lines (though formatting varies):
The world, a tapestry woven from threads of diverse experiences, unfolds each day with a silent promise of novelty, a whisper of the extraordinary hidden within the mundane. From the
first blush of dawn painting the eastern sky to the final twinkling of stars, life moves in rhythms both grand and subtle, a symphony of human endeavor and natural wonder. Consider th
e sprawling metropolis, a hive of ceaseless activity where early risers become a human river flowing to work, a vibrant mosaic of cultures, ambitions, and dreams. The air buzzes with
a thousand conversations, the aroma of street food, and the distant hum of traffic, a concrete jungle alive with interconnected stories. Yet, amidst this chaos, pockets of serenity ex
ist, a quiet park where an elderly couple shares a bench, or a lone artist captures the fleeting beauty of a sunset on canvas.
```

14. tail :

The tail command displays the last few lines of a file, by default it will be showing 10 lines.

Command : **tail <file-name>**

```
prathamesh@Prathamesh-Shingate:~/linux$ tail file3.txt
Then there are the profound moments of human connection, those small, unassuming acts of kindness that ripple outward, transforming a stranger's difficult day into one of hope. A shar
ed smile, a held door, a patient listening ear—these gestures, seemingly insignificant, form the bedrock of our shared humanity, reminding us we are not islands. Think of the child's
wonder discovering a ladybug on a leaf, or the quiet satisfaction of a gardener tending to fragile blooms, each a testament to life's enduring magic. Our journey is also marked by anc
ient whispers from civilizations past, echoed in monumental architecture, stories etched in stone that connect us to millennia of human struggle and triumph, like the enduring mystiqu
e of the Himalayas or the vibrant festivals that celebrate shared heritage. Even in quiet moments of solitude, looking up at the vast, star-dusted expanse, one feels a deep connection
 to something immense, a universal narrative that binds us all. This vast, intricate narrative, encompassing bustling cities, quiet acts of love, and the enduring legacy of history, i
s the essence of our shared existence, a continuous story waiting for each of us to write our own verse. It's a story of resilience, of fleeting beauty, and of the profound, often und
erstated, connections that make life truly meaningful, a ceaseless cycle of beginnings and endings, of joy and sorrow, all woven into the grand, ongoing story of the world.
prathamesh@Prathamesh-Shingate:~/linux$
```

15. df -h

Displays disk space usage of all mounted file systems in a human-readable format**.** Shows disk size, used, available, and mount points

Command **: df -h**

```
prathamesh@Prathamesh-Shingate:~/linux$ df -h
Filesystem      Size  Used Avail Use% Mounted on
tmpfs           1.6G  2.8M  1.6G   1% /run
/dev/nvme0n1p5  144G   19G  117G  14% /
tmpfs           7.7G  422M  7.3G   6% /dev/shm
tmpfs           5.0M  8.0K  5.0M   1% /run/lock
efivarfs        192K  138K   50K  74% /sys/firmware/efi/efivars
/dev/nvme0n1p1  256M   89M  168M  35% /boot/efi
tmpfs           1.6G  3.3M  1.6G   1% /run/user/1000
prathamesh@Prathamesh-Shingate:~/linux$
```

16. free -m:

Display information in Megabytes.

Command : **free -m**

```
prathamesh@Prathamesh-Shingate:~/linux$ free -m
              total        used        free      shared  buff/cache   available
Mem:          15704        4669        8048        1462        4406       11035
Swap:          4095           0        4095
prathamesh@Prathamesh-Shingate:~/linux$
```

17. top :

Provides a real-time view of CPU, memory, and running processes.

Command : **top**

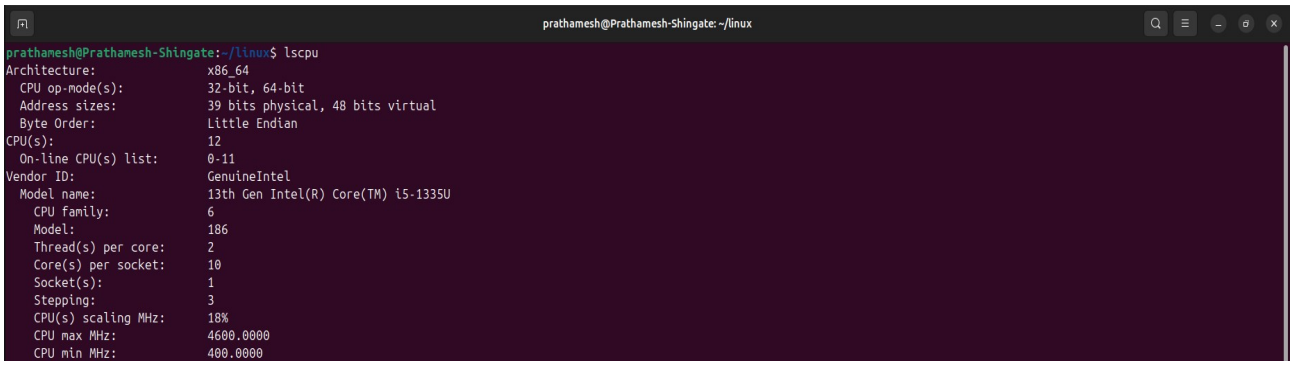
```
prathamesh@Prathamesh-Shingate: ~/linux
top - 22:24:24 up  3:02,  1 user,  load average: 0.51, 0.73, 0.76
Tasks: 322 total,   1 running, 317 sleeping,   2 stopped,   2 zombie
%Cpu(s):  0.7 us,  0.7 sy,  0.0 ni, 98.5 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :  15704.4 total,   8053.4 free,   4662.8 used,   4410.7 buff/cache
MiB Swap:   4096.0 total,   4096.0 free,      0.0 used.  11041.6 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
   2632 pratham+  20   0 5017152 377204 161056 S   9.1   2.3  10:34.89 gnome-shell
      1 root      20   0   23688  14788   9504 S   0.0   0.1   0:03.85 systemd
      2 root      20   0       0      0      0 S   0.0   0.0   0:00.01 kthreadd
      3 root      20   0       0      0      0 S   0.0   0.0   0:00.00 pool_workqueue_release
      4 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-rcu_gp
      5 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-sync_wq
      6 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-kvfree_rcu_reclaim
      7 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-slub_flushwq
      8 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-netns
     10 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/0:0H-events_highpri
     13 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-mm_percpu_wq
     14 root      20   0       0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_kthread
     15 root      20   0       0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_rude_kthread
     16 root      20   0       0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_trace_kthread
     17 root      20   0       0      0      0 S   0.0   0.0   0:00.29 ksoftirqd/0
     18 root      20   0       0      0      0 I   0.0   0.0   0:15.58 rcu_preempt
     19 root      20   0       0      0      0 S   0.0   0.0   0:00.00 rcu_exp_par_gp_kthread_worker/0
     20 root      20   0       0      0      0 S   0.0   0.0   0:00.06 rcu_exp_gp_kthread_worker
     21 root      rt   0       0      0      0 S   0.0   0.0   0:00.42 migration/0
     22 root     -51   0       0      0      0 S   0.0   0.0   0:00.00 idle_inject/0
     23 root      20   0       0      0      0 S   0.0   0.0   0:00.00 cpuhp/0
     24 root      20   0       0      0      0 S   0.0   0.0   0:00.00 cpuhp/2
     25 root     -51   0       0      0      0 S   0.0   0.0   0:00.00 idle_inject/2
     26 root      rt   0       0      0      0 S   0.0   0.0   0:00.55 migration/2
     27 root      20   0       0      0      0 S   0.0   0.0   0:00.24 ksoftirqd/2
     29 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/2:0H-events_highpri
     30 root      20   0       0      0      0 S   0.0   0.0   0:00.00 cpuhp/4
     31 root     -51   0       0      0      0 S   0.0   0.0   0:00.00 idle_inject/4
     32 root      rt   0       0      0      0 S   0.0   0.0   0:03.31 migration/4
     33 root      20   0       0      0      0 S   0.0   0.0   0:00.07 ksoftirqd/4
     35 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/4:0H-events_highpri
     36 root      20   0       0      0      0 S   0.0   0.0   0:00.00 cpuhp/5
     37 root     -51   0       0      0      0 S   0.0   0.0   0:00.00 idle_inject/5
     38 root      rt   0       0      0      0 S   0.0   0.0   0:02.20 migration/5
     39 root      20   0       0      0      0 S   0.0   0.0   0:00.08 ksoftirqd/5
     41 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/5:0H-events_highpri
     42 root      20   0       0      0      0 S   0.0   0.0   0:00.00 cpuhp/6
     43 root     -51   0       0      0      0 S   0.0   0.0   0:00.00 idle_inject/6
     44 root      rt   0       0      0      0 S   0.0   0.0   0:01.45 migration/6
     45 root      20   0       0      0      0 S   0.0   0.0   0:00.05 ksoftirqd/6
     47 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/6:0H-events_highpri
     48 root      20   0       0      0      0 S   0.0   0.0   0:00.00 cpuhp/7
     49 root     -51   0       0      0      0 S   0.0   0.0   0:00.00 idle_inject/7
```

18. lscpu :

Displays detailed information about the CPU architecture, including number of cores, threads, CPU family, model, speed, and architecture type

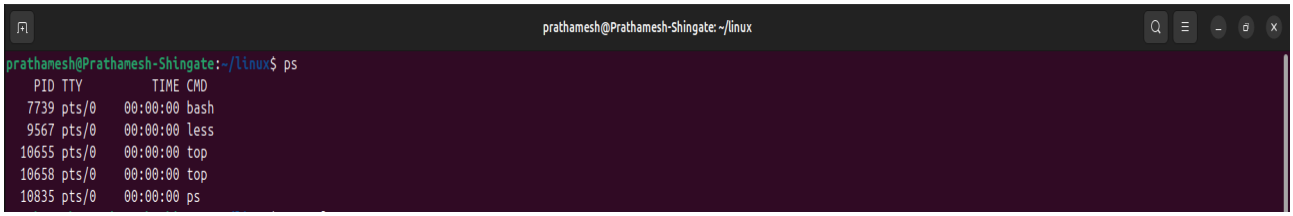
command: **lscpu**



19. ps :

Shows the processes currently running **in your terminal/session.**

Command: **ps**



20. ps -ef :

Displays a full, system-wide list of all running processes, including processes from all users, along with detailed information such as user ID, process ID (PID), parent process ID (PPID), start time.

```
prathamesh@Prathamesh-Shingate:~/linux$ ps -ef
UID          PID    PPID  C STIME TTY          TIME CMD
root           1       0  0 19:21 ?        00:00:03 /sbin/init splash
root           2       0  0 19:21 ?        00:00:00 [kthreadd]
root           3       2  0 19:21 ?        00:00:00 [pool_workqueue_release]
root           4       2  0 19:21 ?        00:00:00 [kworker/R-rcu_gp]
root           5       2  0 19:21 ?        00:00:00 [kworker/R-sync_wq]
root           6       2  0 19:21 ?        00:00:00 [kworker/R-kvfree_rcu_reclaim]
root           7       2  0 19:21 ?        00:00:00 [kworker/R-slub_flushwq]
root           8       2  0 19:21 ?        00:00:00 [kworker/R-netns]
root          10       2  0 19:21 ?        00:00:00 [kworker/0:0H-events_highpri]
root          13       2  0 19:21 ?        00:00:00 [kworker/R-mm_percpu_wq]
root          14       2  0 19:21 ?        00:00:00 [rcu_tasks_kthread]
root          15       2  0 19:21 ?        00:00:00 [rcu_tasks_rude_kthread]
root          16       2  0 19:21 ?        00:00:00 [rcu_tasks_trace_kthread]
root          17       2  0 19:21 ?        00:00:00 [ksoftirqd/0]
root          18       2  0 19:21 ?        00:00:16 [rcu_preempt]
root          19       2  0 19:21 ?        00:00:00 [rcu_exp_par_gp_kthread_worker/0]
root          20       2  0 19:21 ?        00:00:00 [rcu_exp_gp_kthread_worker]
root          21       2  0 19:21 ?        00:00:00 [migration/0]
root          22       2  0 19:21 ?        00:00:00 [idle_inject/0]
root          23       2  0 19:21 ?        00:00:00 [cpuhp/0]
root          24       2  0 19:21 ?        00:00:00 [cpuhp/2]
root          25       2  0 19:21 ?        00:00:00 [idle_inject/2]
```

21. kill :

Sends a signal to a process to control its behavior, most commonly to terminate the process using its process ID (PID).

Command : **kill -9 <process-id>**

```
prathamesh@Prathamesh-Shingate:~/linux$ sleep 1000
^Z
[4]+  Stopped                 sleep 1000
prathamesh@Prathamesh-Shingate:~/linux$ ps
    PID TTY          TIME CMD
   7739 pts/0    00:00:00 bash
   9567 pts/0    00:00:00 less
  10655 pts/0    00:00:00 top
  10658 pts/0    00:00:00 top
  11853 pts/0    00:00:00 sleep
  11895 pts/0    00:00:00 ps
prathamesh@Prathamesh-Shingate:~/linux$ kill -9 11853
prathamesh@Prathamesh-Shingate:~/linux$ ps
    PID TTY          TIME CMD
   7739 pts/0    00:00:00 bash
   9567 pts/0    00:00:00 less
  10655 pts/0    00:00:00 top
  10658 pts/0    00:00:00 top
  11896 pts/0    00:00:00 ps
[4]+  Killed                  sleep 1000
prathamesh@Prathamesh-Shingate:~/linux$
```

22. grep :

Searches for a specific word or pattern inside text or files and displays the matching lines.

Command : **grep <word/pattern> <file-name>**

```
prathamesh@Prathamesh-Shingate:~/linux$ grep "silva" file1.txt
This is silva from dubai
prathamesh@Prathamesh-Shingate:~/linux$
```

23. pipe [|] :

Connects the output of one command directly to the input of another command, allowing commands to work together.

Eq → ps -ef | grep root

This sends the output of ps -ef to grep to show only the lines related to root processes

```
prathamesh@Prathamesh-Shingate:~/linux$ ps -ef | grep root
root           1       0  0 Jan13 ?        00:00:04 /sbin/init splash
root           2       0  0 Jan13 ?        00:00:00 [kthreadd]
root           3       2  0 Jan13 ?        00:00:00 [pool_workqueue_release]
root           4       2  0 Jan13 ?        00:00:00 [kworker/R-rcu_gp]
root           5       2  0 Jan13 ?        00:00:00 [kworker/R-sync_wq]
root           6       2  0 Jan13 ?        00:00:00 [kworker/R-kvfree_rcu_reclaim]
root           7       2  0 Jan13 ?        00:00:00 [kworker/R-slub_flushwq]
root           8       2  0 Jan13 ?        00:00:00 [kworker/R-netns]
root          10       2  0 Jan13 ?        00:00:00 [kworker/0:0H-events_highpri]
root          13       2  0 Jan13 ?        00:00:00 [kworker/R-mm_percpu_wq]
root          14       2  0 Jan13 ?        00:00:00 [rcu_tasks_kthread]
```

24. ifconfig :

Displays network interfaces and their assigned IP addresses.

Command : **ifconfig**

```
prathamesh@Prathamesh-Shingate:~/linux$ ifconfig
docker0: flags=4099<UP,BROADCAST,MULTICAST>  mtu 1500
        inet 172.17.0.1  netmask 255.255.0.0  broadcast 172.17.255.255
        ether 5a:e1:bc:51:69:8b  txqueuelen 0  (Ethernet)
        RX packets 0  bytes 0 (0.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 0  bytes 0 (0.0 B)
        TX errors 0  dropped 46 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 6774  bytes 615199 (615.1 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 6774  bytes 615199 (615.1 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

25.ping :

Checks whether a remote host is reachable over the network or not.

Command : **ping**

eg → ping google.com

```
prathamesh@Prathamesh-Shingate:~/linux$ ping google.com
PING google.com (2404:6800:4009:804::200e) 56 data bytes
64 bytes from bom05s05-in-x0e.1e100.net (2404:6800:4009:804::200e): icmp_seq=1 ttl=115 time=71.4 ms
64 bytes from bom05s05-in-x0e.1e100.net (2404:6800:4009:804::200e): icmp_seq=2 ttl=115 time=101 ms
64 bytes from bom05s05-in-x0e.1e100.net (2404:6800:4009:804::200e): icmp_seq=3 ttl=115 time=45.3 ms
64 bytes from bom05s05-in-x0e.1e100.net (2404:6800:4009:804::200e): icmp_seq=4 ttl=115 time=43.0 ms
64 bytes from bom05s05-in-x0e.1e100.net (2404:6800:4009:804::200e): icmp_seq=5 ttl=115 time=56.2 ms
64 bytes from bom05s05-in-x0e.1e100.net (2404:6800:4009:804::200e): icmp_seq=6 ttl=115 time=88.5 ms
64 bytes from bom05s05-in-x0e.1e100.net (2404:6800:4009:804::200e): icmp_seq=7 ttl=115 time=50.1 ms
```

26. traceroute :

Shows the path that network packets take to reach a destination host.

Eg → **traceroute google.com**

```
prathamesh@Prathamesh-Shingate:~/linux$ traceroute google.com
traceroute to google.com (142.250.195.46), 30 hops max, 60 byte packets
 1  ip-10-8-0-1.ap-south-1.compute.internal (10.8.0.1)  23.689 ms  34.434 ms  34.453 ms
 2  20.20.0.1 (20.20.0.1)  34.451 ms  34.447 ms  34.438 ms
 3  240.2.196.13 (240.2.196.13)  34.475 ms 240.2.196.14 (240.2.196.14)  34.407 ms 240.2.196.13 (240.2.196.13)  34.490 ms
 4  242.6.253.135 (242.6.253.135)  34.369 ms 242.6.253.5 (242.6.253.5)  34.296 ms 242.6.252.3 (242.6.252.3)  34.430 ms
 5  * * *
 6  99.82.178.53 (99.82.178.53)  34.288 ms * *
 7  * * *
 8  142.251.64.14 (142.251.64.14)  168.502 ms 142.251.77.96 (142.251.77.96)  168.454 ms 142.250.227.74 (142.250.227.74)  168.378 ms
 9  192.178.110.248 (192.178.110.248)  168.449 ms 142.250.238.81 (142.250.238.81)  168.246 ms 192.178.110.248 (192.178.110.248)  168.332 ms
10  lcboma-ax-in-f14.1e100.net (142.250.195.46)  168.235 ms 192.178.110.199 (192.178.110.199)  168.234 ms 192.178.110.109 (192.178.110.109)  55.391 ms
prathamesh@Prathamesh-Shingate:~/linux$
```

27. netstat :

Shows active network connections, listening ports, and routing information.

Command : **netstat**

```
prathamesh@Prathamesh-Shingate:~/linux$ netstat
Active Internet connections (w/o servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 ip-10-8-0-22.ap-s:44014 ec2-13-127-99-68.:https CLOSE_WAIT
tcp        0      0 ip-10-8-0-22.ap-s:53976 93.243.107.34.bc.:https ESTABLISHED
tcp        0      0 ip-10-8-0-22.ap-s:59986 ec2-3-108-30-176.:https ESTABLISHED
tcp6       0      0 Prathamesh-Shinga:49330 2a06:98c1:3100::6:https ESTABLISHED
tcp6       0      0 Prathamesh-Shinga:60786 2a06:98c1:3100::6:https ESTABLISHED
tcp6       0      0 Prathamesh-Shinga:50842 2606:4700:90d1:d8:https ESTABLISHED
udp        0      0 ip-10-8-0-22.ap-s:60673 ec2-44-205-171-15:https ESTABLISHED
udp        0      0 ip-10-78-161-159:bootpc ip-10-78-161-151:bootps ESTABLISHED
Active UNIX domain sockets (w/o servers)
Proto RefCnt Flags       Type       State         I-Node   Path
unix  3      [ ]         STREAM     CONNECTED     36778
```

28. nslookup :

Used to check DNS records by translating a domain name into an IP address or an IP address into a domain name.

Command eg → **nslookup google.com**

```
prathamesh@Prathamesh-Shingate:~/linux$ nslookup google.com
Server:         127.0.0.53
Address:        127.0.0.53#53

Non-authoritative answer:
Name:   google.com
Address: 142.250.70.46
Name:   google.com
Address: 2404:6800:4009:813::200e

prathamesh@Prathamesh-Shingate:~/linux$
```

29. du (disk usage) :

Shows disk usage of files and directories.

Command : **du**

```
prathamesh@Prathamesh-Shingate:~$ cd linux
prathamesh@Prathamesh-Shingate:~/linux$ du
4       ./Backup
20      .
prathamesh@Prathamesh-Shingate:~/linux$
```

30. chown :

Changes the ownership of a file or directory from one user to another.

sudo is required because changing ownership needs admin permission.

Command : **sudo chown <user-name> <file/folder>**

```
prathamesh@Prathamesh-Shingate:/$ sudo chown prathamesh file5.txt
prathamesh@Prathamesh-Shingate:/$ ls -al
total 4194404
drwxr-xr-x  23 root       root       4096 Jan 14 00:43 .
drwxr-xr-x  23 root       root       4096 Jan 14 00:43 ..
lrwxrwxrwx   1 root       root          7 Apr 22  2024 bin -> usr/bin
drwxr-xr-x   2 root       root       4096 Feb 26  2024 bin.usr-is-merged
drwxr-xr-x   4 root       root       4096 Jan  2 20:56 boot
drwxr-xr-x   2 root       root       4096 Jan  1  1970 cdrom
drwxr-xr-x  20 root       root       5020 Jan 13 19:21 dev
drwxr-xr-x 143 root       root      12288 Jan 11 17:10 etc
-rw-r--r--   1 prathamesh root          0 Jan 14 00:43 file5.txt
drwxr-xr-x   3 root       root       4096 Jan  2 14:37 home
```

This command makes prathamesh the owner of the file file5.txt.

31. chmod :

Changes file or directory permissions, such as **read, write, or execute** access.

1. **Symbolic mode** is a **human-readable way** to change file permissions using **letters instead of numbers**.

Who (whose permission to change)

u → user (file owner)

g → group

o → others

a → all (user, group, others)

Operator (what action to perform)

+ → add permission

- → remove permission

Permission (what permission)

- **r → read**
- **w → write**
- **x → execute**

eg → chmod +x file1.txt

Adds execute permission to the file, allowing it to be run as a command or script**.**

```
prathamesh@Prathamesh-Shingate:~/linux$ chmod +x file1.txt
prathamesh@Prathamesh-Shingate:~/linux$ ls -al
total 24
drwxrwxr-x  3 prathamesh prathamesh 4096 Jan 13 21:18 .
drwxr-x--- 24 prathamesh prathamesh 4096 Jan 13 20:17 ..
drwxrwxr-x  2 prathamesh prathamesh 4096 Jan 13 20:55 Backup
-rwxrwxr-x  1 prathamesh prathamesh   45 Jan 13 20:29 file1.txt
-rw-rw-r--  1 prathamesh prathamesh 2854 Jan 13 21:04 file2.txt
-rw-rw-r--  1 prathamesh prathamesh 1457 Jan 13 21:18 file3.txt
prathamesh@Prathamesh-Shingate:~/linux$
```

2. Numeric :

sets file permissions using digits for owner, group, and others, where **read = 4, write = 2, execute = 1.**

eg → chmod 644 file6.txt

Owner (6) → rw- (read + write)

Group (4) → r-- (read only)

Others (4) → r-- (read only)

This sets the file so the owner can read/write, and everyone else can only read.

```
prathamesh@Prathamesh-Shingate:~/linux$ chmod 644 file6.txt
prathamesh@Prathamesh-Shingate:~/linux$ ls -al
total 24
drwxrwxr-x  3 prathamesh prathamesh 4096 Jan 14 01:10 .
drwxr-x--- 24 prathamesh prathamesh 4096 Jan 13 20:17 ..
drwxrwxr-x  2 prathamesh prathamesh 4096 Jan 13 20:55 Backup
-rwxrwxr-x  1 prathamesh prathamesh   45 Jan 13 20:29 file1.txt
-rw-r--r--  1 prathamesh prathamesh 2854 Jan 13 21:04 file2.txt
-rw-rw-r--  1 prathamesh prathamesh 1457 Jan 13 21:18 file3.txt
-rw-r--r--  1 prathamesh prathamesh    0 Jan 14 01:10 file6.txt
prathamesh@Prathamesh-Shingate:~/linux$
```

32. man :

used to display the manual or documentation of other commands. It helps you understand a command's syntax, options, and examples.

Command : **man man**

```
MAN(1)                                                    Manual pager utils                                                    MAN(1)

NAME
       man - an interface to the system reference manuals

SYNOPSIS
       man [man options] [[section] page ...] ...
       man -k [apropos options] regexp ...
       man -K [man options] [section] term ...
       man -f [whatis options] page ...
       man -l [man options] file ...
       man -w|-W [man options] page ...

DESCRIPTION
       man  is  the system's manual pager.  Each page argument given to man is normally the name of a program, utility or function.  The manual page associated with each of these
       arguments is then found and displayed.  A section, if provided, will direct man to look only in that section of the manual.  The default action is to search in all of  the
       available sections following a pre-defined order (see DEFAULTS), and to show only the first page found, even if page exists in several sections.

       The table below shows the section numbers of the manual followed by the types of pages they contain.

       1   Executable programs or shell commands
       2   System calls (functions provided by the kernel)
       3   Library calls (functions within program libraries)
       4   Special files (usually found in /dev)
       5   File formats and conventions, e.g. /etc/passwd
       6   Games
       7   Miscellaneous (including macro packages and conventions), e.g. man(7), groff(7), man-pages(7)
       8   System administration commands (usually only for root)
       9   Kernel routines [Non standard]

       A manual page consists of several sections.

       Conventional  section names include NAME, SYNOPSIS, CONFIGURATION, DESCRIPTION, OPTIONS, EXIT STATUS, RETURN VALUE, ERRORS, ENVIRONMENT, FILES, VERSIONS, STANDARDS, NOTES,
       BUGS, EXAMPLE, AUTHORS, and SEE ALSO.

       The following conventions apply to the SYNOPSIS section and can be used as a guide in other sections.

       bold text          type exactly as shown.
       italic text        replace with appropriate argument.
       [-abc]             any or all arguments within [ ] are optional.
       -a|-b              options delimited by | cannot be used together.
       argument ...       argument is repeatable.
       [expression] ...   entire expression within [ ] is repeatable.

       Exact rendering may vary depending on the output device.  For instance, man will usually not be able to render italics when running in a terminal, and will  typically  use
       underlined or coloured text instead.

       The command or function illustration is a pattern that should match all possible invocations.  In some cases it is advisable to illustrate several exclusive invocations as
       is shown in the SYNOPSIS section of this manual page.
```

33. tar :

The tar command is used to collect many files into one archive file.

Eg → tar -cvf backup.tar test/

-c: Create a new archive.

-v: Verbose (shows the progress in the terminal).

-f: Filename (allows you to specify the name of the archive).

```
prathamesh@Prathamesh-Shingate:~/linux$ tar -cvf backup.tar test/
test/
test/new15
test/new13
test/new14
test/new12
prathamesh@Prathamesh-Shingate:~/linux$ ls -al
total 40
drwxrwxr-x  4 prathamesh prathamesh  4096 Jan 14 01:37 .
drwxr-x--- 24 prathamesh prathamesh  4096 Jan 13 20:17 ..
drwxrwxr-x  2 prathamesh prathamesh  4096 Jan 13 20:55 Backup
-rw-rw-r--  1 prathamesh prathamesh 10240 Jan 14 01:37 backup.tar
-rwxrwxr-x  1 prathamesh prathamesh    45 Jan 13 20:29 file1.txt
```

Creating a zipped file :

eg → tar -czf backup.tar.gz test/

c → Create: Start a new archive.

z → Zip: Compress it (make it small).

f → File: file name used

```
prathamesh@Prathamesh-Shingate:~/linux$ tar -czf backup.tar.gz test/
prathamesh@Prathamesh-Shingate:~/linux$ ls -al
total 44
drwxrwxr-x  4 prathamesh prathamesh  4096 Jan 14 01:46 .
drwxr-x--- 24 prathamesh prathamesh  4096 Jan 13 20:17 ..
drwxrwxr-x  2 prathamesh prathamesh  4096 Jan 13 20:55 Backup
-rw-rw-r--  1 prathamesh prathamesh 10240 Jan 14 01:37 backup.tar
-rw-rw-r--  1 prathamesh prathamesh   174 Jan 14 01:46 backup.tar.gz
-rwxrwxr-x  1 prathamesh prathamesh    45 Jan 13 20:29 file1.txt
-rw-r--r--  1 prathamesh prathamesh  2854 Jan 13 21:04 file2.txt
-rw-rw-r--  1 prathamesh prathamesh  1457 Jan 13 21:18 file3.txt
-rw-r--r--  1 prathamesh prathamesh     0 Jan 14 01:10 file6.txt
-rw-rw-r--  1 prathamesh prathamesh     0 Jan 14 01:30 new1
```

2. Extracting Archives :

Extract a compressed archive → **tar -xzvf backup.tar.gz**

-x: xtract the contents.

```
prathamesh@Prathamesh-Shingate:~/linux$ tar -xzvf backup.tar.gz
test/
test/new15
test/new13
test/new14
test/new12
prathamesh@Prathamesh-Shingate:~/linux$ ls -al
total 44
drwxrwxr-x  4 prathamesh prathamesh  4096 Jan 14 01:46 .
drwxr-x--- 24 prathamesh prathamesh  4096 Jan 13 20:17 ..
drwxrwxr-x  2 prathamesh prathamesh  4096 Jan 13 20:55 Backup
-rw-rw-r--  1 prathamesh prathamesh 10240 Jan 14 01:37 backup.tar
-rw-rw-r--  1 prathamesh prathamesh   174 Jan 14 01:46 backup.tar.gz
-rwxrwxr-x  1 prathamesh prathamesh    45 Jan 13 20:29 file1.txt
-rw-r--r--  1 prathamesh prathamesh  2854 Jan 13 21:04 file2.txt
-rw-rw-r--  1 prathamesh prathamesh  1457 Jan 13 21:18 file3.txt
-rw-r--r--  1 prathamesh prathamesh     0 Jan 14 01:10 file6.txt
-rw-rw-r--  1 prathamesh prathamesh     0 Jan 14 01:30 new1
-rw-rw-r--  1 prathamesh prathamesh     0 Jan 14 01:30 new2
-rw-rw-r--  1 prathamesh prathamesh     0 Jan 14 01:30 new3
-rw-rw-r--  1 prathamesh prathamesh     0 Jan 14 01:30 new4
-rw-rw-r--  1 prathamesh prathamesh     0 Jan 14 01:30 new5
drwxrwxr-x  2 prathamesh prathamesh  4096 Jan 14 01:37 test
prathamesh@Prathamesh-Shingate:~/linux$ cd test
prathamesh@Prathamesh-Shingate:~/linux/test$ ls
new12  new13  new14  new15
prathamesh@Prathamesh-Shingate:~/linux/test$
```